Core-Dimension Sequences Need Not Be Eventually Recurrent

Literature-Already-Answered Identification

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Status. `literature_already_answered`. This is an exact later negative answer, not a new counterexample produced by the run.

1 Source question

For a finite group G, a field k of characteristic p, and a finite dimensional kG-module M, let

$$c_n^G(M) = \dim_k \operatorname{core}(M^{\otimes n}),$$

where the core is the direct sum of the nonprojective indecomposable summands. Question 5.15.5 on source-PDF page 111 of Benson's memoir asks whether, for every M, the numbers $c_n^G(M)$ satisfy a constant-coefficient linear recurrence for all sufficiently large n [1]. Equivalently, it asks whether $\sum_{n\geq 0} c_n^G(M)t^n$ is always rational.

The memoir notes that this is among the questions originating in the earlier work of Benson and Symonds. Meng formulates their same general question as the stronger Question 13.3 behind his Question 4.20 [2].

2 Later negative answer

Meng's Theorem 4.21 (supporting-PDF page 25) constructs Ω-algebraic modules for which $c_n^G(M)$ is not eventually recursive. The concrete example immediately following the theorem is

$$G = V_4, \qquad M = \Omega(k) \oplus \Omega^{-1}(k).$$

Thus a finite group and finite-dimensional module violate the universal claim in Benson's Question 5.15.5.

The supporting paper gives two equivalent checks. First, its asymptotic theorem yields

$$c_n^G(M) \sim C\, 2^n n^{1/2}$$

on the relevant parity classes, with $C \neq 0$. Coefficients of a rational generating function are finite sums of exponential terms times polynomials of *integer* degree on residue classes; the factor $n^{1/2}$ is incompatible with that form. Second, Example 5.3 displays the core series

$$(1+2z)\left(\frac{2}{1-4z^2} + \frac{4}{(1-4z^2)^{3/2}}\right),$$

which is algebraic but not rational. Hence the answer to the source question is no.

3 Verification and scope

Theorem 4.21 is stated for a wider family. Its proof uses nonzero variance of the syzygy-index distribution; the displayed V_4 example has indices $+1$ and -1 with equal weights and therefore has variance one. The concrete counterexample is consequently within the proof's noninteger-exponent case, independently of any zero-variance edge case in the broad theorem wording.

This identification settles only Question 5.15.5. It does not answer the memoir's adjacent questions about $\gamma_G(M \otimes M^*)$, symmetry of the completed representation ring, the possible values of $\gamma_G(M) < 2$, or algebraic integrality.

A bounded search used the exact memoir title, the phrases "linear recurrence relation with constant coefficients" and "eventually recursive", and the Benson–Symonds invariant/core-series terminology. It found Meng's 2026 paper, which explicitly labels the relevant Benson–Symonds question and states that Theorem 4.21 answers it negatively.

References

[1] D. J. Benson, *Modular representation theory and commutative Banach algebras*, Memoirs of the American Mathematical Society 298 (2024), no. 1488; arXiv:2008.13155.

[2] C. Meng, *Asymptotic behavior of modular representations over abelian p-groups*, arXiv:2603.11592, 2026.